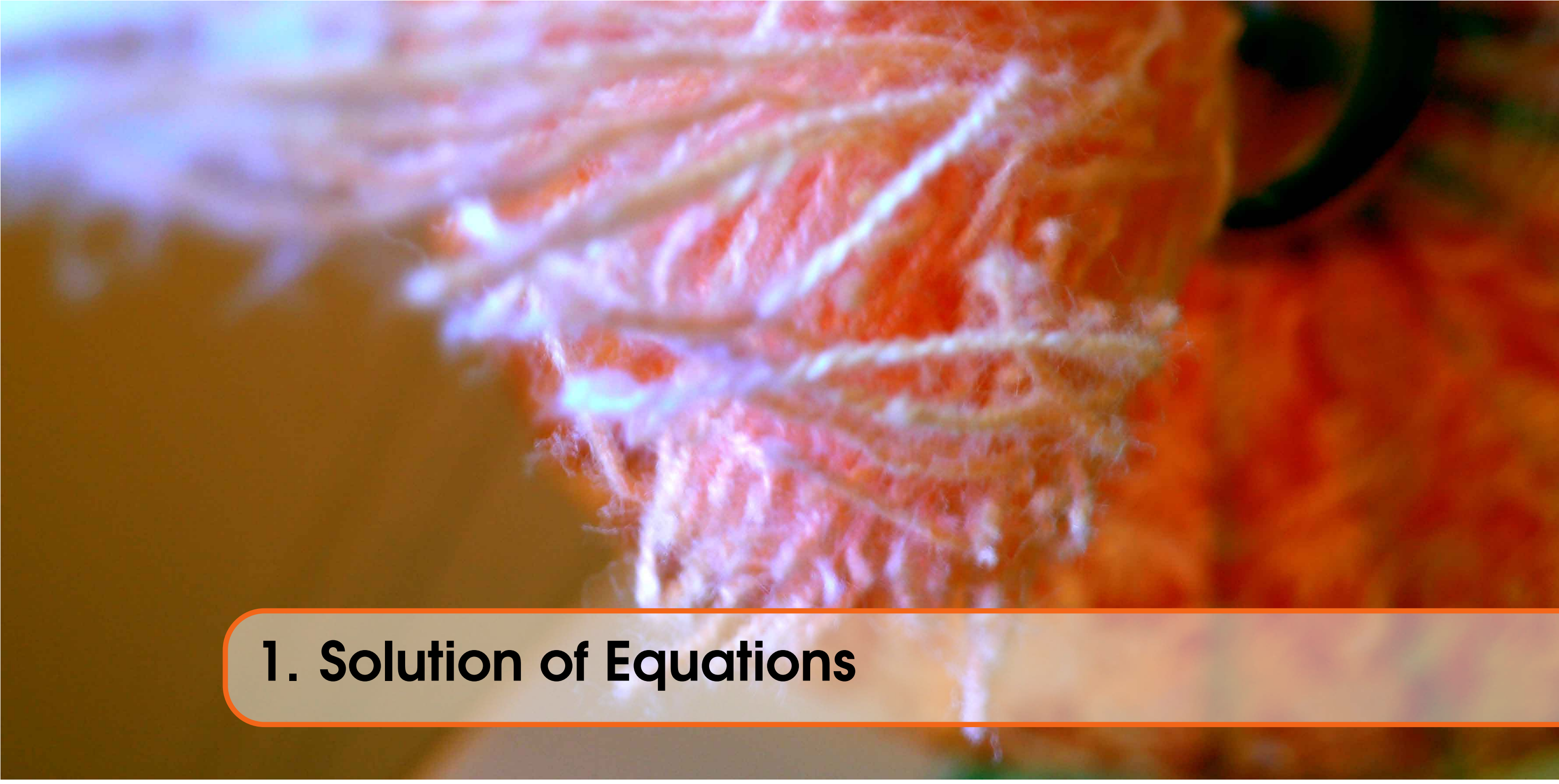
Module I



1. Solution of Equations

1.1 Solution of Algebraic and Transcendental Equations

1.1.1 Introduction

A problem of great importance in science and engineering is that of determining the roots/zeros of an equation of the form

$$f(x) = 0, \tag{1.1}$$

A polynomial equation of the form

$$f(x) = P_n(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n = 0 \tag{1.2}$$

is called an *algebraic equation*. An equation which contains polynomials, exponential functions,logarithmic functions, trigonometric functions etc. is called a *transcendental equation*.

For example,

$$3x^3 - 2x^2 - x - 5 = 0, \qquad x^4 - 3x^2 + 1 = 0, \qquad x^2 - 3x + 1 = 0,$$

are algebraic (polynomial) equations, and

$$xe^{2x} - 1 = 0, \qquad \cos x - xe^x = 0, \qquad \tan x = x$$

are transcendental equations.

We assume that the function $f(x)$ is continuous in the required interval.

We define the following.

Definition 1.1.1 — Root/zero. A number α, for which $f(\alpha) \equiv 0$ is called a root of the equation $f(x) = 0$, or a zero of $f(x)$. Geometrically, a root of an equation $f(x) = 0$ is the value of x at which the graph of the equation $y = f(x)$ intersects the x-axis (see Fig. 1.1).

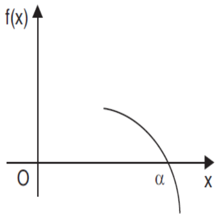


Figure 1.1: Root of $f(x) = 0$

Definition 1.1.2 — Simple root. A number α is a simple root of $f(x) = 0$, if $f(\alpha) = 0$ and $f'(\alpha) \neq 0$. Then, we can write $f(x)$ as

$$f(x) = (x-\alpha)g(x), g(\alpha) \neq 0. \tag{1.3}$$

For example, since $(x-1)$ is a factor of $f(x) = x^3 + x - 2 = 0$, we can write

$$f(x) = (x-1)(x^2+x+2) = (x-1)g(x), g(1)\neg 0.$$

Alternately, we find $f(1) = 0$, $f'(x) = 3x^2 + 1$, $f'(1) = 4\neg 0$. Hence, $x = 1$ is a simple root of

$$f(x) = x^3 + x - 2 = 0.$$

Definition 1.1.3 — Multiple root. A number α is a multiple root, of multiplicity m, of $f(x) = 0$, if

$$f(\alpha) = 0, f'(\alpha) = 0, ..., f^{n-1}(\alpha) = 0, and f^m(\alpha)\neg 0. \tag{1.4}$$

Then, we can write $f(x)$ as

$$f(x) = (x-\alpha)^m g(x), g(\alpha)\neg 0.$$

For example, consider the equation $f(x) = x^3 - 3x^2 + 4 = 0$. We find

$$f(2) = 8 - 12 + 4 = 0, f'(x) = 3x^2 - 6x, f'(2) = 12 - 12 = 0,$$
$$f'\imath(x) = 6x - 6, f'\imath = 6\neg 0.$$

Hence, $x = 2$ is a multiple root of multiplicity 2 (double root) of $f(x) = x^3 - 3x^2 + 4 = 0$.

We can write $\qquad f(x) = (x-2)^2(x+1) = (x-2)^2 g(x), g(2) = 3 \neq 0.$

In this chapter, we shall be considering the case of simple roots only.

R A polynomial equation of degree n has exactly n roots, real or complex, simple or multiple, where as a transcendental equation may have one root, infinite number of roots or no root.

We shall derive methods for finding only the real roots.

The methods for finding the roots are classified as (i) direct methods, and (ii) iterative methods.

Definition 1.1.4 — Direct methods. These methods give the exact values of all the roots in a finite number of steps (disregarding the round-off errors). Therefore, for any direct method, we can give the total number of operations (additions, subtractions, divisions and multiplications). This number is called the *operational count* of the method.

For example, the roots of the quadratic equation $ax^2+bx+c=0$, $a\neg 0$, can be obtained using the method

$$x=\frac{1}{2a}\left[-b\pm\sqrt{b^2-4ac}\right].$$

For this method, we can give the count of the total number of operations.
There are direct methods for finding all the roots of cubic and fourth degree polynomials. However, these methods are difficult to use.
Direct methods for finding the roots of polynomial equations of degree greater than 4 or transcendental equations are not available in literature.

Definition 1.1.5 — Iterative methods. These methods are based on the idea of successive approximations. We start with one or two initial approximations to the root and obtain a sequence of approximations $x0, x1, \ldots, x_k, \ldots$, which in the limit as $k\to\infty$, converge to the exact root α. An iterative method for finding a root of the equation $f(x)=0$ can be obtained as

$$x_{k+1}=\phi(x_k), k=0,1,2,\ldots \tag{1.5}$$

This method uses one initial approximation to the root .0x The sequence of approximations is given by

$$x_1=\phi(x_0), x_2=\phi(x_1), x_3=\phi(x_2),\ldots$$

The function ϕ is called an *iteration function* and x_0 is called an *initial approximation*.

If a method uses two initial approximations x_0, x_1, to the root, then we can write the method as

$$x_{k+1}=\phi(xk-1, x_k), k=1,2,\ldots \tag{1.6}$$

Definition 1.1.6 — Convergence of iterative methods. The sequence of iterates, x_k, is said to converge to the exact root α, if

$$\lim_{k\to\infty} x_k=\alpha, or \lim_{k\to\infty}[|x_k-\alpha|]=0. \tag{1.7}$$

The error of approximation at the k th iterate is defined as $\varepsilon_k=x_k-\alpha$. Then, we can write (1.7) as

$$\lim_{k\to\infty} \mid \text{ error of approximation } \Big| = \lim_{k\to\infty}\Big| x_k-\alpha \Big| = \lim_{k\to\infty}\Big| \varepsilon_k \mid = 0.$$

R Given one or two initial approximations to the root, we require a suitable iteration function ϕ for a given function $f(x)$, such that the sequence of iterates, x_k, converge to the exact root α. Further, we also require a suitable criterion to terminate the iteration.

Definition 1.1.7 — Criterion to terminate iteration procedure. Since, we cannot perform infinite number of iterations, we need a criterion to stop the iterations. We use one or both of the following criterion:

(i) The equation $f(x)=0$ is satisfied to a given accuracy or $f(x_k)$ is bounded by an *error tolerance* ε.

$$|f(x_k)|\le\varepsilon \tag{1.8}$$

(ii) The magnitude of the difference between two successive iterates is smaller than a given accuracy or an error bound ε.

$$|x_{k+1} - x_k| \leq \varepsilon. \tag{1.9}$$

Generally, we use the second criterion. In some very special problems, we require to use both the criteria.
For example, if we require two decimal place accuracy, then we iterate until $|x_{k+1} - x_k| \leq 0.005$. If we require three decimal place accuracy, then we iterate until $|x_{k+1} - x_k| \leq 0.0005$.

As we have seen earlier, we require a suitable iteration function and suitable initial approximation(s) to start the iteration procedure. In the next section, we give a method to find initial approximation(s).

1.1.2 Initial Approximation for an Iterative Procedure

For polynomial equations, *Descartes' rule of signs* gives the bound for the number of positive and negative real roots.

(i) We count the number of changes of signs in the coefficients of $P_n(x)$ for the equation $f(x) = P_n(x) = 0$. The number of positive roots cannot exceed the number of changes of signs. For example, if there are four changes in signs, then the equation may have four positive roots or two positive roots or no positive root. If there are three changes in signs, then the equation may have three positive roots or definitely one positive root. (For polynomial equations with real coefficients, complex roots occur in conjugate pairs.)

(ii) We write the equation $f(-x) = P_n(-x) = 0$, and count the number of changes of signs in the coefficients of $P_n(-x)$. The number of negative roots cannot exceed the number of changes of signs. Again, if there are four changes in signs, then the equation may have four negative roots or two negative roots or no negative root. If there are three changes in signs, then the equation may have three negative roots or definitely one negative root.

We use the following theorem of calculus to determine an initial approximation. It is also called the *intermediate value theorem*.

Theorem 1.1.1 If $f(x)$ is continuous on some interval $[a,b]$ and $f(a)f(b) \leq 0$, then the equation $f(x) = 0$ has at least one real root or an odd number of real roots in the interval (a,b).

This result is very simple to use. We set up a table of values of $f(x)$ for various values of x. Studying the changes in signs in the values of $f(x)$, we determine the intervals in which the roots lie. For example, if $f(1)$ and $f(2)$ are of opposite signs, then there is a root in the interval $(1,2)$.

Let us illustrate through the following examples.

Problem 1.1 Determine the maximum number of positive and negative roots and intervals of length one unit in which the real roots lie for the following equations.

(i) $8x^3 - 12x^2 - 2x + 3 = 0$

(ii) $3x^3 = 2x^2 - x - 5 = 0$.

Solution. (i) Let $f(x) = 8x^3 - 12x^2 - 2x + 3 = 0$.
The number of changes in the signs of the coefficients $(8, -12, -2, 3)$ is 2. Therefore, the equation has 2 or no positive roots. Now, $f(-x) = -8x^3 - 12x^2 + 2x + 3$. The number of changes in signs in the coefficients $(-8, -12, 2, 3)$ is 1. Therefore, the equation has one negative root.We have the following table of values for $f(x)$.

x	-2	-1	0	1	2	3
$f(x)$	-105	-15	3	-3	15	105

Table 1.1

Since,

$$f(-1)f(0) < 0, \text{ there is a root in the interval } (-1,0),$$
$$f(0)f(1) < 0, \text{ there is a root in the interval } (0,1),$$
$$f(1)f(2) < 0, \text{ there is a root in the interval } (1,2).$$

Therefore, there are three real roots and the roots lie in the intervals (- 1, 0), (0, 1), (1, 2).
(ii) Let $f(x) = 3x^3 - 2x^2 - x - 5 = 0$.
The number of changes in the signs of the coefficients $(3, -2, -1, -5)$ is 1. Therefore,the equation has one positive root. Now, $f(-x) = -3x^2 - 2x^2 + x - 5$.The number of changes in signs in the coefficients $(-3, -2, 1, -5)$ is 2. Therefore, the equation has two negative or no negative roots.

We have the table of values for $f(x)$.

x	-3	-2	-1	0	1	2	3
$f(x)$	-101	-35	-9	-5	-5	9	55

Table 1.2

From the table, we find that there is one real positive root in the interval $(1,2)$. The equation has no negative real root. ∎

Problem 1.2 Determine an interval of length one unit in which the negative real root, which is smallest in magnitude lies for the equation $9x^3 + 18x^2 - 37x - 70 = 0$.

Solution. Let $f(x) = 9x^3 + 18x^2 - 37x - 70 = 0$. Since, the smallest negative real root in magnitude is required, we form a table of values for $x < 0$, .

x	-5	-4	-3	-2	-1	0
$f(x)$	-560	-210	-40	4	-24	-70

Table 1.3

Since, $f(-2)f(-1) < 0$, the negative root of smallest magnitude lies in the interval $(-2, -1)$ ∎

Problem 1.3 Locate the smallest positive root of the equations

(i) $xe^x = \cos x$
(ii) $\tan x = 2x$

Solution. (i) Let $f(x) = xe^x - \cos x = 0$. We have $f(0) = -1, f(1) = e - \cos 1 = 2.718 - 0.540 = 2.178$. Since, $f(0)f(1) < 0$, there is a root in the interval $(0,1)$.

(ii) Let $f(x) = \tan x - 2x = 0$. We have the following function values.

$$f(0) = 0, f(0.1) = -0.0997, f(0.5) = -0.4537,$$
$$f(1) = -0.4426, f(1,1) = -0.2352, f(1.2) = 0.1722.$$

Since, $f(1.1)f(1.2) < 0$, the root lies in the interval $(1.1, 1.2)$.

Now, we present some iterative methods for finding a root of the given algebraic or transcendental equation.

We know from calculus, that in the neighborhood of a point on a curve, the curve can be approximated by a straight line. For deriving numerical methods to find a root of an equation

$f(x) = 0$, we approximate the curve in a sufficiently small interval which contains the root, by a straight line. That is, in the neighborhood of a root, we approximate

$$f(x) \approx ax + b, \quad a \neq 0$$

where a and b are arbitrary parameters to be determined by prescribing two appropriate conditions on $f(x)$ and/or its derivatives. Setting $ax+b = 0$, we get the next approximation to the root as $x = -b/a$. Different ways of approximating the curve by a straight line give different methods. These methods are also called chord methods. Method of false position (also called regula-falsi method) and Newton-Raphson method fall in this category of chord methods. ■

1.2 Bisection Method

This method is based on the repeated application of the intermediate value theorem. If we know that root of $f(x) = 0$ lies in the interval $I_0 = (a_0, b_0)$, we bisect I_0 at the point $m_1 = (a_0 + b_0)/2$. Denote by I_1 the interval (a_0, m_1) if $f(a_0)f(m_1) < 0$ or the interval (m_1, b_0) if $f(m_1)f(b_0) < 0$. Therefore, the interval I_1 also contains the root. We bisect the interval I_1 and get a subinterval I_2 at whose end points $f(x)$ takes the values of opposite signs and therefore contains the root. Continuing this procedure, we obtain a sequence of nested sets of sub-intervals $I_0 \supset I_1 \supset I_2 \supset \cdots$ such that each subinterval contains the root. After repeating the bisecting process q times, we either find the root or find the interval I_q of length $(b_0 - a_0)/2^q$ which contains the root. We take the midpoint of the last subinterval as the desired approximation to the root.

This root has error not greater than one-half of the length of the interval of which it is the midpoint. Thus, we have

$$m_{k+1} = a_k + \frac{1}{2}(b_k - a_k), k = 0, 1, 2, \cdots$$

where

$$(a_{k+1}, b_{k+1}) = \begin{cases} (a_k, m_{k+1}), & if\ f(a_k)f(m_{k+1}) < 0 \\ (m_{k+1}, b_k), & if\ f(m_{k+1})f(b_k) < 0 \end{cases}$$

We notice that this method uses only the end points of the interval $[a_k, b_k]$ for which $f(a_k)f(b_k) < 0$ and not the values of $f(x)$ at these end points, to obtain the next approximation to the root.

1.2.1 Solved Problems

Problem 1.4 Perform five iterations of the bisecting method to obtain the smallest positive root of the equation

$$f(x) = x^3 - 5x + 1.$$

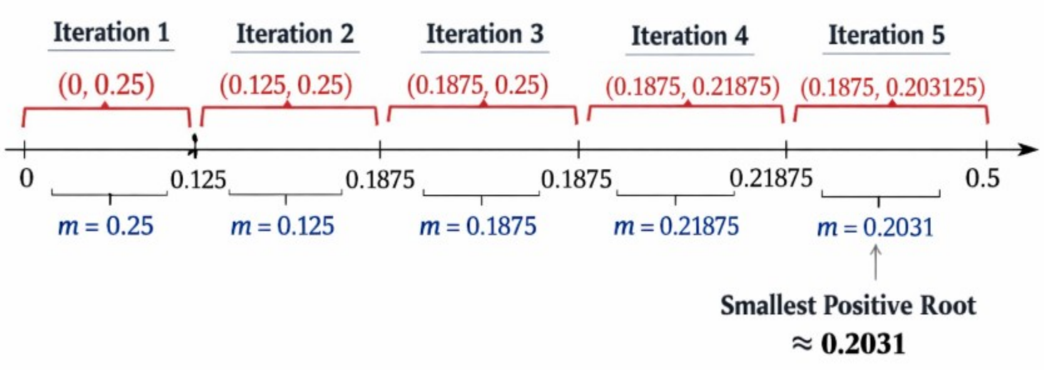
Solution. Since $f(0) > 0$ and $f(1) < 0$, the smallest positive root lies in the interval $(0, 1)$. Taking $a_0 = 0$, $b_0 = 1$, we get

$$m_1 = \frac{1}{2}(a_0 + b_0) = \frac{1}{2}(0 + 1) = 0.5$$
$$f(m_1) = -1.375 \text{ and } f(a_0)f(m_1) < 0.$$

Thus, the root lies in the interval $(0, 0.5)$. Taking $a_1 = 0$, $b_1 = 0.5$, we get

$$m_2 = \frac{1}{2}(a_1 + b_1) = \frac{1}{2}(0 + 0.5) = 0.25$$
$$f(m_2) = f(0.25) = -0.234375 \text{ and } f(a_1)f(m_2) < 0.$$

a	a_{k-1}	b_{k-1}	m_k	$f(m_k)$	$f(m_k)f(a_{k-1})$
1	0	1	0.5	-1.375	<0
2	0	0.5	0.25	-0.2344	<0
3	0	0.25	0.125	0.377	>0
4	0.125	0.25	0.1875	0.0691	>0
5	0.1875	0.25	0.21875	-0.0833	<0



Thus, the root lies in the interval $(0, 0.25)$. The sequence of intervals is given in the following table. Hence, the root lies in $(0.1875, 0.21875)$. The approximate root is taken as the midpoint of this interval, that is 0.203125.

■

1.2.2 Practice Problems:

1. Using Bisecting method find the real roots of the equation $f(x) = x^3 - x - 1 = 0$.
2. Using Bisecting method find the real roots of the equation $x^3 - 2x - 5 = 0$.
3. Using Bisecting method find the real roots of the equation $f(x) = x^3 + x^2 + x + 7 = 0$, correct the three decimal places.
4. Find a root, correct to three decimal places and lying between o and 0.5, of the equation $4e^{-x}\sin x - 1 = 0$

1.3 Secant Method

This method is an improvement over the method of false position as it does not require the condition $f(x_0)f(x_1) < 0$ of that method (Fig. 2.5). Here also the graph of the function $y = f(x)$ is approximated by a secant line but at each iteration, two most recent approximations to the root are used to find the next approximation. Also it is not necessary that the interval must contain the root.

Taking x_0, x_1 as the initial limits of the interval, we write the equation of the chord joining these as

$$y - f(x_1) = \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_1)$$

Then the abscissa of the point where it crosses the x-axis $(y = 0)$ is given by

$$x_2 = x_1 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_1)$$

which is an approximation to the root. The general formula for successive approximations is, therefore, given by

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n), n \geq 1$$

1.3.1 Rate of Convergence

If at any iteration $f(x_n) = f(x_{n-1})$, this method fails and shows that it does not converge necessarily. This is a drawback of secant method over the method of false position which always converges. But if the secant method once converges, its rate of convergence is 1.6 which is faster than that of the method of false position.

Problem 1.5 Find a root of the equation $x^3 - 2x - 5 = 0$ using secant method correct to three decimal places.

Solution. Let $f(x) = x^3 - 2x - 5$ so that

$$
\begin{aligned}
f(0) &= (0)^3 - 2(0) - 5 = -5 = -ve\\
f(1) &= (1)^3 - 2(1) - 5 = -6 = -ve\\
f(2) &= (2)^3 - 2(2) - 5 = -1 = -ve\\
f(3) &= (3)^3 - 2(3) - 5 = 16 = +ve
\end{aligned}
$$

$\therefore$ Taking initial approximations $x_0 = 2$ and $x_1 = 3$, by secant method, we have

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

So we get,

$$x_2 = x_1 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_1) = 3 - \frac{3-2}{16+1} 16 = 2.058823$$

Now $f(x_2) = -0.390799$

$$\therefore \quad x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2) = 2.081263$$

and

$$f(x_3) = -0.147204$$

Therefore,

$$x_4 = x_3 - \frac{x_3 - x_2}{f(x_3) - f(x_2)} f(x_3) = 2.094824$$

and

$$f(x_4) = 0.003042$$

Therefore,

$$x_5 = x_4 - \frac{x_4 - x_3}{f(x_4) - f(x_3)} f(x_4) = 2.094549$$

Hence the root is 2.095 correct to 3 decimal places. ■

Problem 1.6 Find the root of the equation $xe^x = \cos x$ using the secant method correct to four decimal places.

Solution. Let $f(x) = \cos x - xe^x = 0$. So that

$$f(0) = \cos 0 - (0)e^{(0)} = 1 = +ve$$

$$f(1) = \cos(1) - (1)e^{(1)} = 0.5403 - 2.7183 = -2.17798 = -ve$$

Taking the initial approximations $x_0 = 0, x_1 = 1$.

Then by secant method, we have

$$x_2 = x_1 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_1) = 1 + \frac{1}{3.17798}(-2.17798) = 0.31467$$

Now $f(x_2) = 0.51987$

Therefore

$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2) = 0.44673$$

and

$$f(x_3) = 0.20354$$

Therefore

$$x_4 = x_3 - \frac{x_3 - x_2}{f(x_3) - f(x_2)} f(x_3) = 0.53171.$$

Repeating this process, the successive approximations are $x_5 = 0.51690, x_6 = 0.51775$, $x_7 = 0.51776$ etc.

Hence the root is 0.5178 correct to 4 decimal places. ■

1.4 Newton-Raphson Method

This method is also called Newton's method. This method is also a chord method in which we approximate the curve near a root, by a straight line.

Let x_0 be an initial approximation to the root of $f(x) = 0$. Then, $P(x_0, f_0)$, where $f_0 = f(x_0)$, is a point on the curve. Draw the tangent to the curve in the neighborhood of the root by the tangent to the curve at the point P, Figure 1.2. We approximate the curve in the neighborhood of the root by the tangent to the curve at the point P. The point of intersection of the tangent with the x-axis is taken as the next approximation to the root. The process is repeated until the required accuracy is obtained. The equation of the tangent to the curve $y = f(x)$ at the point $P(x_0, f_0)$ is given by

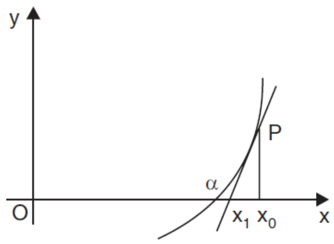


Figure 1.2: 'Newton-Raphson method'

$$y - f(x_0) = (x - x_0) f'(x_0)$$

where $f'(x_0)$ is the slope of the tangent to the curve at P. Setting $y = 0$ and solving for x, we get

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}, \quad f'(x_0) \neq 0$$

The next approximation to the root is given by

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}, \quad f'(x_0) \neq 0$$

We repeat the procedure. The iteration method is defined as

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad f'(x_k) \neq 0 \tag{1.10}$$

This method is called the Newton-Raphson method or simply the Newton's method. The method is also called the *tangent method.*

Alternate derivation of the method

Let x_k be an approximation to the root of the equation $f(x) = 0$. Let Δx be an increment in x such that $x_k + \Delta x$ is the exact root, that is $f(x_k + \Delta x) \equiv 0$.
Expanding in Taylor's series about the point x_k, we get

$$f(x_k) + \Delta x f'(x_k) + \frac{(\Delta x)^2}{2!} f''(x_k) + \ldots = 0. \tag{1.11}$$

Neglecting the second and higher powers of Δx, we obtain

$$f(x_k) + \Delta x f'(x_k) \approx 0 \text{or} \Delta x = -\frac{f(x_k)}{f'(x_k)}.$$

Hence, we obtain the iteration method

$$x_{k+1} = x_k + \Delta x = x_k - \frac{f(x_k)}{f'(x_k)}, \quad f'(x_k) \neq 0, \quad k = 0, 1, 2, \ldots$$

which is same as the method derived earlier.

R Convergence of the Newton's method depends on the initial approximation to the root. If the approximation is far away from the exact root, the method diverges (see Example 1.6). However, if a root lies in a small interval (a, b) and $x_0 \varepsilon (a, b)$, then the method converges.

R From equation (1.10), we observe that the method may fail when $f'(x)$ is close to zero in the neighborhood of the root. Later, in this section, we shall give the condition for convergence of the method.

R The computational cost of the method is one evaluation of the function $f(x)$ and one evaluation of the derivative $f'(x)$, for each iteration.

Problem 1.7 Derive the Newton's method for finding $1/N$, where $N > 0$. Hence, find $1/17$, using the initial approximation as (i) 0.05, (ii) 0.15. Do the iterations converge ?

Solution.

Let $$x = \frac{1}{N},$$

or $$\frac{1}{x} = N.$$

Define $$f(x) = \frac{1}{x} - N.$$

Then, $$f'(x) = -\frac{1}{x^2}.$$

Newton's method gives

$$x = x_k - \frac{f(x_k)}{f'(x_k)} = x_k - \frac{[(1/x_k) - N]}{[-1/x_k^2]} = x_k + [x_k - N x_k^2] = 2x_k - N x_k^2$$

.

(i) With N = 17, and $x_0 = 0.05$, we obtain the sequence of approximations

$$x_1 = 2x_0 - Nx_0^2 = 2(0.05) - 17(0.05)^2 = 0.0575.$$
$$x_2 = 2x_1 - Nx_1^2 = 2(0.0575) - 17(0.0575)^2 = 0.058794.$$
$$x_3 = 2x_2 - Nx_2^2 = 2(0.058794) - 17(0.058794)^2 = 0.058823.$$
$$x_4 = 2x_1 = 3 - Nx_3^2 = 2(0.058823) - 17(0.058823)^2 = 0.058823.$$

Since, $|x_4 - x_3| = 0$, the iterations converge to the root. The required root is 0.058823.

(ii) With $N = 17$, and $x_0 = 0.15$, we obtain the sequence of approximations

$$x_1 = 2x_0 - Nx_0^2 = 2(0.15) - 17(0.15)^2 = -0.0825.$$
$$x_2 = 2x_1 - Nx_1^2 = 2(-0.0825) - 17(-0.0825)^2 = -0.280706.$$
$$x_3 = 2x_2 - Nx_2^2 = 2(-0.280706) - 17(-0.280706)^2 = -1.900942.$$
$$x_4 = 2x_3 - Nx_3^2 = 2(-1.900942) - 17(-1.900942)^2 = -65.23275.$$

We find that $x_k \to -\infty$ as k increases. Therefore, the iterations diverge very fast. This shows the importance of choosing a proper initial approximation.

■

Problem 1.8 Derive the Newton's method for finding the q^{th} root of a positive number $N, N^{1/q}$, where $N > 0, q > 0$. Hence, compute $17^{1/3}$ correct to four decimal places, assuming the initial approximation as $x_0 = 2$.

Solution.

Let	$x = N^{1/q}$,
or	$x^q = N$.
Define	$f(x) = x^q - N$.
Then,	$f'(x) = qx^{q-1}$.

Newton's method gives the iteration

$$x_{k+1} = x_k - \frac{x_k^q - N}{qx_k^{q-1}} = \frac{qx_k^q - x_k^q + N}{qx_k^{q-1}} = \frac{(q-1)x_k^q + N}{qx_k^{q-1}}.$$

For computing $17^{1/3}$, we have $q = 3$ and $N = 17$. Hence, the method becomes

$$x_{k+1} = \frac{2x_k^3 + 17}{3x_k^2}, \quad k = 0, 1, 2, \ldots$$

With $x_0 = 2$, we obtain the following results.

$$x_1 = \frac{2x_0^3 + 17}{3x_0^2} = \frac{2(8) + 17}{3(4)} = 2.75,$$
$$x_2 = \frac{2x_1^3 + 17}{3x_1^2} = \frac{2(2.75)^3 + 17}{3(2.75)^2} = 2.582645$$
$$x_3 = \frac{2x_2^3 + 17}{3x_2^2} = \frac{2(2.582645)^3 + 17}{3(2.582645)^2} = 2.571332$$
$$x_4 = \frac{2x_3^3 + 17}{3x_3^2} = \frac{2(2.571332)^3 + 17}{3(2.571332)^2} = 2.571282$$

Now, $|x_4 - x_3| = |2.571282 - 2.571332| = 0.00005$
We may take $x \approx 2.571282$ as the required root correct to four decimal places. ■

Problem 1.9 Perform four iterations of the Newton's method to find the smallest positive root of the equation $f(x) = x^3 - 5x + 1 = 0$.

Solution. We have $f(0) = 1, f(1) = -3$. Since, $f(0)f(1) < 0$, the smallest positive root lies in the interval $(0,1)$.

Define $\quad f(x) = x_k^3 - 5x_k + 1.$

Then, $\quad f'(x) = 3x_k^2 - 5$.

Applying the Newton's method, we obtain

$$x_{k+1} = x_k - \frac{x_k^3 - 5x_k + 1}{3x_k^2 - 5} = \frac{2x_k^3 - 1}{3x_k^2 - 5}, \quad k = 0,1,2,\ldots$$

Let $x_0 = 0.5$. We have the following results.

$$x_1 = \frac{2x_0^3 - 1}{3x_0^2 - 5} = \frac{2(0.5)^3 - 1}{3(0.5)^2 - 5} = 0.176471,$$
$$x_2 = \frac{2x_1^3 - 1}{3x_1^2 - 5} = \frac{2(0.176471)^3 - 1}{3(0.176471)^2 - 5} = 0.201568,$$
$$x_3 = \frac{2x_2^3 - 1}{3x_2^2 - 5} = \frac{2(0.201568)^3 - 1}{3(0.201568)^2 - 5} = 0.201640,$$
$$x_4 = \frac{2x_3^3 - 1}{3x_3^2 - 5} = \frac{2(0.201640)^3 - 1}{3(0.201640)^2 - 5} = 0.201640.$$

Therefore, the root correct to six decimal places is $x \approx 0.201640$. ■

Problem 1.10 Using Newton-Raphson method solve $x \log_{10} x = 12.34$ with $x_0 = 10$.

Solution.

Define $\quad f(x) = x\log_{10} x - 12.34.$

Then $\quad f'(x) = \log_{10} x + \dfrac{1}{\log_e 10} = \log_{10} x + 0.434294.$

Using the Newton-Raphson method, we obtain

$$x_{k+1} = x_k - \frac{x_k \log_{10} x_k - 12.34}{\log_{10} x_k + 0.434294}, k = 0,1,2,\ldots$$

With $x_0 = 10$, we obtain the following results.

$$x_1 = x_0 - \frac{x_0 \log_{10} x_0 - 12.34}{\log_{10} x_0 + 0.434294} = 10 - \frac{10\log_{10} 10 - 12.34}{\log_{10} 10 + 0.434294} = 11.631465.$$
$$x_2 = x_1 - \frac{x_1 \log_{10} x_1 - 12.34}{\log_{10} x_1 + 0.434294}$$
$$= 11.631465 - \frac{11.631465\log_{10} 11.631465 - 12.34}{\log_{10} 11.631465 + 0.434294} = 11.594870.$$
$$x_3 = x_2 - \frac{x_2 \log_{10} x_2 - 12.34}{\log_{10} x_2 + 0.434294}$$
$$= 11.59487 - \frac{11.59487\log_{10} 11.59487 - 12.34}{\log_{10} 11.59487 + 0.434294} = 11.594854.$$

We have $|x_3 - x_2| = |11.594854 - 11.594870| = 0.000016$.
We may take $x \approx 11.594854$ as the root correct to four decimal places. ■

1.5 Fixed Point Iteration Method or General Iteration Method or Method of Successive Approximations

The method is also called iteration method or method of successive approximations or fixed point iteration method.

The first step in this method is to rewrite the given equation $f(x) = 0$ in an equivalent form as

$$x = \phi(x). \tag{1.12}$$

There are many ways of rewriting $f(x) = 0$ in this form.

For example, $f(x) = x^3 - 5x + 1 = 0$, can be rewritten in the following forms.

$$x = \frac{x^3+1}{5}, x = (5x-1)^{1/3}, x = \sqrt{\frac{5x-1}{x}}, \text{ etc.} \tag{1.13}$$

Now, finding a root of $f(x) = 0$ is same as finding a number α such that $\alpha = \phi(\alpha)$, that is, a fixed point of $\phi(x)$. A fixed point of a function ϕ is a point α such that $\alpha = \phi(\alpha)$. This result is also called the fixed point theorem.

Using equation (1.12), the iteration method is written as

$$x_{k+1} = \phi(x_k), k = 0, 1, 2, \ldots \tag{1.14}$$

The function $\phi(x)$ is called the iteration function. Starting with the initial approximation x_0, we compute the next approximations as

$$x_1 = \phi(x_0), x_2 = \phi(x_1), x_3 = \phi(x_2), \ldots$$

The stopping criterion is same as used earlier. Since, there are many ways of writing $f(x) = 0$ as $x = \phi(x)$, it is important to know whether all or at least one of these iteration methods converges.

R Convergence of an iteration method $x_{k+1} = \phi(x_k), k = 0, 1, 2, \ldots$, depends on the choice of the iteration function $\phi(x)$, and a suitable initial approximation x_0, to the root.

Consider again, the iteration methods given in equation (1.13), for finding a root of the equation $f(x) = x^3 - 5x + 1 = 0$. The positive root lies in the interval $(0, 1)$.

(i)

$$x_{k+1} = \frac{x_k^3+1}{5}, k = 0, 1, 2, \ldots \tag{1.15}$$

With $x_0 = 1$, we get the sequence of approximations as

$$x_1 = 0.4, x_2 = 0.2128, x_3 = 0.20193, x_4 = 0.20165, x_5 = 0.20164.$$

The method converges and $x \approx x_5 = 0.20164$ is taken as the required approximation to the root.

(ii)

$$x_{k+1} = (5x_k - 1)^{1/3}, k = 0, 1, 2, \ldots \tag{1.16}$$

With $x_0 = 1$, we get the sequence of approximations as

$$x_1 = 1.5874, x_2 = 1.9072, x_3 = 2.0437, x_4 = 2.0968, \ldots$$

which does not converge to the root in $(0, 1)$.

(iii) $x_{k+1}=\sqrt{\frac{5x_k-1}{x_k}}, k=0,1,2,\ldots$ With $x_0=1$, we get the sequence of approximations as

$$x_1=2.0, x_2=2.1213, x_3=2.1280, x_4=2.1284, \ldots \tag{1.17}$$

which does not converge to the root in $(0,1)$.

Now, we derive the condition that the iteration function $\phi(x)$ should satisfy in order that the method converges.

Condition of convergence

The iteration method for finding a root of $f(x)=0$, is written as

$$x_{k+1}=\phi\left(x_k\right), k=0,1,2,\ldots \tag{1.18}$$

Let α be the exact root. That is,

$$\alpha=\phi(\alpha). \tag{1.19}$$

We define the error of approximation at the k th iterate as $\varepsilon_k=x_k-\alpha, k=0,1,2,\ldots$

Subtracting (1.19) from (1.18), we obtain

$$\begin{aligned} x_{k+1}-\alpha &= \phi\left(x_k\right)-\phi(\alpha) \\ &= \left(x_k-\alpha\right)\phi'\left(t_k\right) \quad \text{(using the mean value theorem)} \end{aligned} \tag{1.20}$$

or $$\varepsilon_{k+1}=\phi'\left(t_k\right)\varepsilon_k, x_k<t_k<\alpha.$$

Setting $k=k-1$, we get $\varepsilon_k=\phi'\left(t_{k-1}\right)\varepsilon_{k-1}, \quad x_{k-1}<t_{k-1}<\alpha.$

Hence, $$\varepsilon_{k+1}=\phi'\left(t_k\right)\phi'\left(t_{k-1}\right)\varepsilon_{k-1}.$$

Using equation (1.20) recursively, we get

$$\varepsilon_{k+1}=\phi'\left(t_k\right)\phi'\left(t_{k-1}\right)\ldots\phi'\left(t_0\right)\varepsilon_0.$$

The initial error ε_0 is known and is a constant. We have

$$\left|\varepsilon_{k+1}\right|=\left|\phi'\left(t_k\right)\right|\left|\phi'\left(t_{k-1}\right)\right|\ldots\left|\phi'\left(t_0\right)\right|\left|\varepsilon_0\right|.$$

Let $$\left|\phi'\left(t_k\right)\right|\leq c, \quad k=0,1,2,\ldots$$

Then, $$\left|\varepsilon_{k+1}\right|\leq c^{k+1}\left|\varepsilon_0\right|. \tag{1.21}$$

For convergence, we require that $\left|\varepsilon_{k+1}\right|\to 0$ as $k\to\infty$. This result is possible, if and only if $c<1$. Therefore, the iteration method (1.22) converges, if and only if

$$\left|\phi'\left(x_k\right)\right|\leq c<1, k=0,1,2,\ldots$$

or $$\left|\phi'(x)\right|\leq c<1, \text{ for all } x \text{ in the interval } (a,b). \tag{1.22}$$

We can test this condition using x_0, the initial approximation, before the computations are done.

Let us now check whether the methods (1.15), (1.16), (1.17) converge to a root in $(0,1)$ of the equation $f(x)=x^3-5x+1=0$.

(i) We have $\phi(x)=\frac{x^3+1}{5}, \phi'(x)=\frac{3x^2}{5}$, and $\left|\phi'(x)\right|=\frac{3x^2}{5}\leq 1$ for all x in $0<x<1$. Hence, the method converges to a root in $(0,1)$.

(ii) We have $\phi(x) = (5x-1)^{1/3}, \phi'(x) = \frac{5}{3(5x-1)^{2/3}}$. Now $|\phi'(x)| < 1$, when x is close to 1 and $|\phi'(x)| > 1$ in the other part of the interval. Convergence is not guaranteed.

(iii) We have $\phi(x) = \sqrt{\frac{5x-1}{x}}, \phi'(x) = \frac{1}{2x^{3/2}(5x-1)^{1/2}}$. Again, $|\phi'(x)| < 1$, when x is close to 1 and $|\phi'(x)| > 1$ in the other part of the interval. Convergence is not guaranteed.

R Sometimes, it may not be possible to find a suitable iteration function $\phi(x)$ by manipulating the given function $f(x)$. Then, we may use the following procedure. Write $f(x) = 0$ as $x = x + \alpha f(x) = \phi(x)$, where α is a constant to be determined. Let x_0 be an initial approximation contained in the interval in which the root lies. For convergence, we require

$$\left|\phi'(x_0)\right| = \left|1 + \alpha f'(x_0)\right| < 1. \tag{1.23}$$

Simplifying, we find the interval in which α lies. We choose a value for α from this interval and compute the approximations. A judicious choice of a value in this interval may give faster convergence.

Problem 1.11 Find the smallest positive root of the equation $x^3 - x - 10 = 0$, using the general iteration method.

Solution. We have

$$f(x) = x^3 - x - 10, f(0) = -10, f(1) = -10,$$
$$f(2) = 8 - 2 - 10 = -4, f(3) = 27 - 3 - 10 = 14.$$

Since, $f(2)f(3) < 0$, the smallest positive root lies in the interval $(2,3)$.

Write $x^3 = x + 10$, and $x = (x+10)^{1/3} = \phi(x)$. We define the iteration method as

$$x_{k+1} = (x_k + 10)^{1/3}.$$

We obtain $\phi'(x) = \frac{1}{3(x+10)^{2/3}}$.

We find $|\phi'(x)| < 1$ for all x in the interval $(2,3)$. Hence, the iteration converges.

Let $x_0 = 2.5$. We obtain the following results.

$$x_1 = (12.5)^{1/3} = 2.3208, x_2 = (12.3208)^{1/3} = 2.3097,$$
$$x_3 = (12.3097)^{1/3} = 2.3090, x_4 = (12.3090)^{1/3} = 2.3089.$$

Since, $|x_4 - x_3| = 2.3089 - 2.3090 \,| = 0.0001$, we take the required root as $x \approx 2.3089$. ■

Problem 1.12 Find the smallest negative root in magnitude of the equation $3x^4 + x^3 + 12x + 4 = 0$, using the method of successive approximations.

Solution. We have

$$f(x) = 3x^4 + x^3 + 12x + 4 = 0, f(0) = 4, f(-1) = 3 - 1 - 12 + 4 = -6$$

Since, $f(-1)f(0) < 0$, the smallest negative root in magnitude lies in the interval (- 1, 0). Write the given equation as

$$x(3x^3 + x^2 + 12) + 4 = 0, \text{ and } x = -\frac{4}{3x^3 + x^2 + 12} = \phi(x)$$

Write the given equation as

$$x(3x^3 + x^2 + 12) + 4 = 0, \text{and} x = -\frac{4}{3x^3 + x^2 + 12} = \phi(x)$$

The iteration method is written as

$$x_{k+1} = -\frac{4}{3x_k^3 + x_k^2 + 12}.$$

We obtain
$$\phi\prime(x) = \frac{4(9x^2 + 2x)}{3x_k^3 + x_k^2 + 12}.$$

We find $|\phi\prime(x)| < 1$ for all x in the interval (-1,0). Hence, the iteration converges.
Let $x_0 = -0.25$. We obtain the following results.

$$x_1 = -\frac{4}{3(-0.25)^3 + (-0.25)^2 + 12} = -0.33290$$
$$x_2 = -\frac{4}{3(-0.3329)^3 + (-0.3329)^2 + 12} = -0.33333$$
$$x_3 = -\frac{4}{3(-0.33333)^3 + (-0.33333)^2 + 12} = -0.33333$$

The required approximation to the root is $x \approx -0.33333$. ■

Problem 1.13 The equation $f(x) = 3x^3 + 4x^2 + 4x + 1 = 0$ has a root in the interval $(-1,0)$. Determine an iteration function $\phi(x)$, such that the sequence of iterations obtained from $x_{k+1} = \phi(x_k), x_0 = -0.5, k = 0, 1, ...,$ converges to the root.

Solution. We illustrate the method given in Remark 10. We write the given equation as

$$x = x + \alpha(3x^3 + 4x^2 + 4x + 1) = \phi(x)$$

where α is a constant to be determined such that

$$|\phi\prime(x)| = \left|1 + \alpha f'(x_0)\right| = \left|1 + \alpha(9x^2 + 8x + 4)\right| < 1$$

for all $x\varepsilon(-1,0)$. This condition is also to be satisfied at the initial approximation. Setting $x_0 = -0.5$ we get

$$\left|\phi'(x_0)\right| = \left|1 + \alpha f'(x_0)\right| = \left|1 + \frac{9\alpha}{4}\right| < 1$$

or $\quad -1 < 1 + \frac{9\alpha}{4} < 1$ or $\quad -\frac{8}{9} < \alpha < 0$

Hence, α takes negative values. The interval for α depends on the initial approximation x_0. Let us choose the value $\alpha = -0.5$. We obtain the iteration method as

$$\begin{aligned} x_{k+1} &= x_k - 0.5(3x_k^3 + 4x_k^2 + x_k + 1) \\ &= -0.5(3x_k^3 + 4x_k^2 + 2x_k + 1) = \phi(x_k). \end{aligned}$$

Starting with $x_0 = -0.5$, we obtain the following results.

$$
\begin{aligned}
x_1 &= \phi(x_0) = -0.5(3x_0^3 + 4x_0^2 + 2x_0 + 1) \\
&= -0.5[3(-0.5)^3 + 4(-0.5)^2 + 2(-0.5) + 1] = -0.3125. \\
x_2 &= \phi(x_1) = -0.5(3x_1^3 + 4x_1^2 + 2x_1 + 1) \\
&= -0.5[3(-0.3125)^3 + 4(-0.3125)^2 + 2(-0.3125) + 1] = -0.337036. \\
x_3 &= \phi(x_2) = -0.5(3x_2^3 + 4x_2^2 + 2x_2 + 1) \\
&= -0.5[3(-0.337036)^3 + 4(-0.337036)^2 + 2(-0.337036) + 1] = -0.332723. \\
x_4 &= \phi(x_3) = -0.5(3x_3^3 + 4x_3^2 + 2x_3 + 1) \\
&= -0.5[3(-0.332723)^3 + 4(-0.332723)^2 + 2(-0.332723) + 1] = -0.333435. \\
x_5 &= \phi(x_4) = -0.5(3x_4^3 + 4x_4^2 + 2x_4 + 1) \\
&= -0.5[3(-0.333435)^3 + 4(-0.333435)^2 + 2(-0.333435) + 1] = -0.333316.
\end{aligned}
$$

Since $|x_5 - x_4| = |-0.333316 + 0.333435| = 0.000119 < 0.0005$ the result is correct to three decimal places.

We can take the approximation as $x \approx x_5 = -0.333316$. The exact root is $x = -1/3$.

We can verify that $\left|\phi'(x_j)\right| < 1$ for all j. ■